

Finite-volume scalar spectral constructive Gibbs measure – integrated referee package

TECT verification-first repository · claim A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE

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1. Purpose and scope

This package gives a single-file proof and reproduction map for the finite-volume real-scalar constructive measure in Sector A. It consolidates the previously approved analytic proof and scoped T6 enactment without adding a theorem or enlarging the scope. The torus, regulator, covariance, interaction, topology of convergence, and admissible observables are all fixed below so that the result can be attacked without consulting a sibling note.

The underlying theorem was promoted to T6 after Jusang Lee independently ran the hash-bound verifier on 2026-07-18. Jusang Lee confirmed integrated v2.0 on 2026-07-19, but final adversarial review then found that its shell notation did not isolate the zero mode when $q_0 = 0$. The current v2.1 repairs that endpoint without changing the theorem. It is the new exact entry candidate for the eventual reproduction bundle. Repository policy requires renewed operator confirmation of this document before packaging; until that confirmation, no A4 PUBLISHED bundle is claimed.

The result is deliberately scalar. It is not a construction of the full three-complex-component derivative Class-II measure and it proves neither an infinite-volume phase transition nor BCC existence or selection.

1 Definitions and theorem

Let

$$\mathbb{T}_L^3 = [0, L)^3, \quad L = 16, \quad \alpha = \frac{2\pi}{L}, \quad |\mathbb{T}_L^3| = L^3. \quad (2.1)$$

Use the orthonormal real Fourier representation, with the usual conjugacy constraint, and let P_N retain precisely the modes $n \in \mathbb{Z}^3$ satisfying $|n|_\infty \leq N$. For

$$m_{\text{sh}}^2 > 0, \quad Y > 0, \quad q_0 \geq 0, \quad (2.2)$$

define

$$K_n = m_{\text{sh}}^2 + Y(\alpha^2 |n|^2 - q_0^2)^2, \quad C_n = K_n^{-1}. \quad (2.3)$$

Let μ_0 be the centred real Gaussian probability measure with covariance $C = K^{-1}$. Its projected law is $\mu_{0,N} = (P_N)_\# \mu_0$.

For $\lambda \in \mathbb{R}$ and $\gamma > 0$, set

$$V(\phi) = \int_{\mathbb{T}_L^3} \left(\frac{\lambda}{4} \phi(x)^4 + \frac{\gamma}{6} \phi(x)^6 \right) dx. \quad (2.4)$$

The finite-dimensional Galerkin Gibbs law is

$$d\nu_N(\phi_N) = Z_N^{-1} e^{-V(\phi_N)} d\mu_{0,N}(\phi_N). \quad (2.5)$$

Theorem (finite-volume scalar spectral constructive measure). Under (2.2) and $\lambda \in \mathbb{R}$, $\gamma > 0$:

1. $C = K^{-1}$ is trace class on real $L^2(\mathbb{T}_L^3)$ and defines a Radon Gaussian probability measure μ_0 ;
2. $V(\phi)$ is finite for μ_0 -almost every ϕ , every Z_N is finite and strictly positive, and the bounds are uniform in N ;
3. with

$$d\nu(\phi) = Z^{-1} e^{-V(\phi)} d\mu_0(\phi), \quad (2.6)$$

the full sequence ν_N converges weakly to ν on real L^2 ;

4. on the common Gaussian space, the lifted normalized densities converge in $L^1(\mu_0)$, equivalently in total variation; and
5. expectations of every finite-degree polynomial in finitely many continuous real linear functionals on L^2 converge.

This is a T6 conditional theorem under the named hypotheses **A1-SHELL-POSITIVITY** and **A2-H2-SEXTIC-COERCIVITY**. The hard analytic dependency is **A1-SCALAR-ANALYTIC-BRANCH**.

2 Trace class and the sharp Gaussian Hilbert threshold

For every integer $m \geq 1$, the number of lattice points on the nonzero max-norm shell $|n|_\infty = m$ is exactly

$$(2m+1)^3 - (2m-1)^3 = 24m^2 + 2. \quad (3.1)$$

Let

$$m_0 = \max \left\{ 1, \left\lceil \sqrt{2}q_0/\alpha \right\rceil \right\}.$$

If $m \geq m_0$ and $|n|_\infty = m$, then $\alpha^2 |n|^2 \geq 2q_0^2$ and

$$K_n \geq \frac{Y}{4} \alpha^4 m^4. \quad (3.2)$$

Therefore, for $M \geq m_0$,

$$\sum_{|n|_\infty \geq M} C_n \leq \frac{4}{Y\alpha^4} \sum_{m=M}^{\infty} (24m^{-2} + 2m^{-4}) < \infty. \quad (3.3)$$

The inner modes are finite in number and $K_n \geq m_{\text{sh}}^2$, so $\text{Tr } C < \infty$. Hence μ_0 is a Radon Gaussian measure on L^2 .

With the H^s weight, the shell contribution is asymptotic to m^{2s-2} . Thus the covariance is trace class on H^s for every $s < 1/2$. Conversely, the upper estimate $K_n \leq c(1 + |n|^4)$ at large $|n|$ gives the harmonic lower behaviour at $s = 1/2$, so the threshold is sharp. This regularity fact is used as a warning, not as a Sobolev route to the local sextic term.

3 The local interaction: direct Gaussian L^6 control

Let $X_N = (I - P_N)\phi$. Translation invariance gives the pointwise variance

$$t_N = |\mathbb{T}_L^3|^{-1} \sum_{|n|_\infty > N} C_n \longrightarrow 0. \quad (4.1)$$

For a centred real Gaussian random variable X , $\mathbb{E}X^6 = 15(\mathbb{E}X^2)^3$. Tonelli's theorem therefore yields

$$\mathbb{E} \int_{\mathbb{T}_L^3} |X_N(x)|^6 dx = 15 |\mathbb{T}_L^3| t_N^3 \longrightarrow 0. \quad (4.2)$$

Consequently $P_N \phi \rightarrow \phi$ in $L^6(\Omega \times \mathbb{T}_L^3)$, $\phi \in L^6$ almost surely, and the quartic and sextic integrals in (2.4) are finite. This is the load-bearing construction of V ; the false embedding $H^{1/2-} \hookrightarrow L^6$ in three dimensions is never used.

The polynomial difference identity and Holder's inequality on the finite measure product space imply

$$\mathbb{E}|V(P_N \phi) - V(\phi)| \longrightarrow 0. \quad (4.3)$$

For the sextic part one uses

$$|a^6 - b^6| \leq 6(|a|^5 + |b|^5)|a - b| \quad (4.4)$$

with the L^6 convergence and uniform Gaussian L^6 norms; the quartic term follows identically with lower powers.

4 Stability and uniform partition bounds

Put $a = \max(-\lambda, 0)$ and $x = \phi^2$. Direct minimization on $x \geq 0$ gives

$$\frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6 \geq -\frac{a^3}{12\gamma^2} =: -c_*. \quad (5.1)$$

Therefore

$$0 < e^{-V(P_N \phi)} \leq e^{c_* |\mathbb{T}_L^3|}, \quad Z_N \leq e^{c_* |\mathbb{T}_L^3|}. \quad (5.2)$$

Let $\sigma_*^2 = |\mathbb{T}_L^3|^{-1} \text{Tr } C$ and $\lambda_+ = \max(\lambda, 0)$. The real Gaussian moments $\mathbb{E}X^4 = 3\sigma^4$ and $\mathbb{E}X^6 = 15\sigma^6$ give

$$\mathbb{E}V(P_N \phi) \leq |\mathbb{T}_L^3| \left(\frac{3\lambda_+}{4} \sigma_*^4 + \frac{5\gamma}{2} \sigma_*^6 \right) =: B_* < \infty. \quad (5.3)$$

Since $x \mapsto e^{-x}$ is convex, Jensen's inequality has the direction

$$Z_N = \mathbb{E}e^{-V(P_N \phi)} \geq e^{-\mathbb{E}V(P_N \phi)} \geq e^{-B_*} > 0. \quad (5.4)$$

Equations (5.2)–(5.4) enclose every Galerkin partition function uniformly.

5 Full-sequence cutoff removal

Write $w_N = e^{-V(P_N \phi)}$ and $w = e^{-V(\phi)}$. Equation (4.3) implies $V(P_N \phi) \rightarrow V(\phi)$ in probability. The deterministic ceiling in (5.2) makes the weights uniformly bounded, and bounded convergence in probability gives

$$w_N \longrightarrow w \quad \text{in } L^1(\mu_0), \quad Z_N = \mathbb{E}w_N \longrightarrow Z = \mathbb{E}w > 0. \quad (6.1)$$

Define the lifted law

$$d\tilde{\nu}_N = Z_N^{-1} w_N d\mu_0. \quad (6.2)$$

Then

$$\left\| \frac{w_N}{Z_N} - \frac{w}{Z} \right\|_{L^1(\mu_0)} \longrightarrow 0. \quad (6.3)$$

This is total-variation convergence only on the common Gaussian space.

The actual Galerkin law satisfies $\nu_N = (P_N)_\# \tilde{\nu}_N$. For bounded continuous F on L^2 , $P_N \phi \rightarrow \phi$ almost surely and (5.2)–(5.4) uniformly bound the normalized densities. Combining this fact with (6.3) gives

$$\int F(\phi_N) d\nu_N = \int F(P_N \phi) d\tilde{\nu}_N \longrightarrow \int F(\phi) d\nu. \quad (6.4)$$

Thus the full sequence converges weakly; no unnamed subsequence remains. The embedded finite-dimensional laws have different supports and are not claimed to converge in total variation.

6 Declared correlations

Let $\ell_1, \dots, \ell_r$ be continuous real linear functionals on L^2 and Q a finite-degree polynomial. The random variables $Q(\ell_1(P_N \phi), \dots, \ell_r(P_N \phi))$ converge in every finite Gaussian moment to the corresponding unprojected polynomial. The uniform density ceiling following from (5.2)–(5.4) transfers uniform integrability to the lifted Gibbs laws. Hence

$$\int Q(\ell_1(\phi_N), \dots, \ell_r(\phi_N)) d\nu_N \longrightarrow \int Q(\ell_1(\phi), \dots, \ell_r(\phi)) d\nu. \quad (7.1)$$

Unsmearing coincident-point products and composite-operator renormalization are not included.

7 Reproduction and independent execution

The one-command verifier is

```
python codes/foundations/a4_scalar_constructive_measure_verify.py
```

and must print

```
PASS: primary (18/18)
PASS: independent (15/15)
ASSERTS: 33/33
A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS
```

The primary audit derives the trace, stability, partition, and Gaussian-tail bounds. The second route is non-importing: it reconstructs the shell count, Fourier traces, Decimal potential minimum, Gaussian moments, and partition bounds with scalar loops. The integrated verifier launches both routes in a fresh temporary directory and checks the three source hashes pinned in `constructive_measure_manifest.json`.

Jusang Lee independently executed verifier v1.0.0 from clean commit

```
7eee2fe84887cc4ccdd311c75095ec84bc9d0d45
```

on 2026-07-18. The immutable operator artifact is

```
claims/A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE/runs/
2026-07-18-jusang-independent/result.json
```

with SHA-256

```
fb2d81ba75d95972e55c675146fbf84cb8209847ea031abdaa0a5f44b111935c
```

and 31/31 assertions, zero failures, zero bad source rows, and zero bad audit rows. Verifier v1.1.0 changed no mathematics or audit calculation; it removed only the stale pre-promotion boundary sentence. Final review of v2.0 then identified the $q_0 = 0$ shell-index endpoint described above. Primary and independent audit v1.1.0 add separate boundary assertions, and verifier v1.2.0 reruns the full package. Its v2.1 referee preflight passes 33/33, at

claims/A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE/runs/
2026-07-19-referee-preflight-v2.1/result.json

with SHA-256

85da0df0d2b96dbfc98f2ea8a0787bf1bd711228505c671c06cb9d3e036836d8.

The deterministic internal and operator JSONs may have identical content hashes; execution independence is instead evidenced by the clean source commit, separate operator-created path and filesystem event, and fresh subprocess return codes. Algorithmic diversity comes from the non-importing second route.

8 Quantitative sanity record

At cutoff 8 the scalar-loop and vectorized covariance traces agree to at most 1.32×10^{-16} relative error. The production-local shell mass is reconstructed from its stored inputs as

$$r - \frac{Z^2}{4Y} = 0.260000000009475, \quad (9.1)$$

consistent with the stored value 0.26 at the declared source precision. The conservative covariance-trace bounds are 3029.399163006103 for the perturbative scalar anchor and 596.857766699839 for the production-local anchor. The pinned-coefficient pointwise stability constant is 0.0025246087994716246 per unit volume. These numbers are executable sanity checks, not physical predictions or practical cutoff error estimates.

9 Devil’s-advocate review

1. **Trace class alone defines the local sextic. UPHOLD as false.** The separate Gaussian sixth-moment estimate (4.2) is load-bearing.
2. **The proof silently uses $H^{1/2-} \hookrightarrow L^6$. UPHOLD as false.** That embedding is unavailable in three dimensions; the proof uses direct $L^6(\Omega \times \mathbb{T}^3)$ convergence.
3. **A negative quartic makes the Gibbs weight uncontrollable. DISMISSED under the hypotheses.** The positive sextic gives the exact global lower bound (5.1).
4. **Jensen’s inequality has the wrong direction. DISMISSED.** Convexity of e^{-x} gives the lower bound (5.4).
5. **Tightness identifies only a subsequence. UPHOLD as insufficient by itself.** The common-space L^1 density limit identifies the full sequence before weak convergence is concluded.
6. **The finite-dimensional laws converge in total variation after zero extension. UPHOLD as false.** Total variation is asserted only for the lifted laws; the projected laws converge weakly.
7. **Identical deterministic JSON hashes prove a copied operator artifact. DISMISSED.** The execution provenance and subprocess returns are distinct; a non-importing second algorithm supplies diversity.
8. **The theorem constructs derivative Class-II currents. UPHOLD as false.** Typical Gaussian regularity does not automatically define those terms; they require a separate construction or renormalization.

9. **Finite volume proves a phase transition or BCC selection. UPHELD as false.** Neither a thermodynamic limit nor a selection argument occurs here.
10. **The large explicit constants are usable numerical error bars. UPHELD as false.** They certify integrability and reproduction only.
11. **The shell-count proof is undefined at $q_0 = 0$. DISMISSED after v2.1 repair.** The zero Fourier mode is part of the finite inner set, while the tail begins at $m_0 = \max\{1, \lceil \sqrt{2q_0/\alpha} \rceil\}$; two executable routes test the $q_0 = 0, m_0 = 1$ endpoint independently.

10 Falsifier and honesty boundary

The theorem fails if the covariance trace diverges in the declared positive q^4 scope; the Gaussian L^6 tail does not vanish; the exact stability or uniform partition bounds fail; the common-space densities do not converge in L^1 ; two cutoff subsequences have different weak limits; a declared smeared cylinder polynomial fails to converge; a source/evidence hash drifts; or one of the two executable routes fails.

The theorem excludes the full three-component derivative Class-II Gibbs measure, $\eta_{\text{shell}} \neq 0$, zero or negative shell mass, unsmeared composite operators, infinite volume, phase transition, finite-difference Route B and regulator independence, BCC existence or selection, T7, and any claim that the scalar branch shares the full-production parameter anchor.

Result footer

Result ID:	A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE
Precise statement:	Full sharp spectral Galerkin Gibbs sequence converges weakly on real L2; lifted densities converge in L1/TV; finite-degree smeared cylinder polynomial correlations converge.
Scope:	T6 CONDITIONAL-THEOREM@FINITE-VOLUME-REAL-SCALAR-SPECTRAL; L=16, m_sh^2>0, Y>0, q0>=0, lambda real, gamma>0, beta=1.
Dependencies:	A1-SCALAR-ANALYTIC-BRANCH; named hypotheses A1-SHELL-POSITIVITY and A2-H2-SEXTIC-COERCIVITY.
Evidence grade:	ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/a4_scalar_constructive_measure_verify.py
Expected output:	primary 18/18; independent 15/15; total 33/33; A4-SCALAR-CONSTRUCTIVE-INTEGRATED-PASS.
Operator evidence:	2026-07-18-jusang-independent/result.json; SHA-256 fb2d81ba75d95972e55c675146fbf84c...
Publication preflight:	2026-07-19-referee-preflight-v2.1/result.json; SHA-256 85da0df0d2b96dbfc98f2ea8a0787bf1...
Falsification gate:	Failed proof step, route, source/evidence hash, or execution provenance in the declared scope.
Tier:	T6 CONDITIONAL-THEOREM, operator-approved 2026-07-18.
Package status:	exact v2.1 entry document awaits renewed operator confirmation after the q0=0 endpoint repair; no PUBLISHED bundle exists before that confirmation.
No-overclaim statement:	Not derivative Class-II, unsmeared composites, infinite volume, phase transition, Route B, BCC, T7, or parameter identity with the full branch.
Next required action:	Operator review/confirmation of this exact v2.1; then build, integrity-check, and register the T6 PUBLISHED bundle before the A5 capstone bundle.